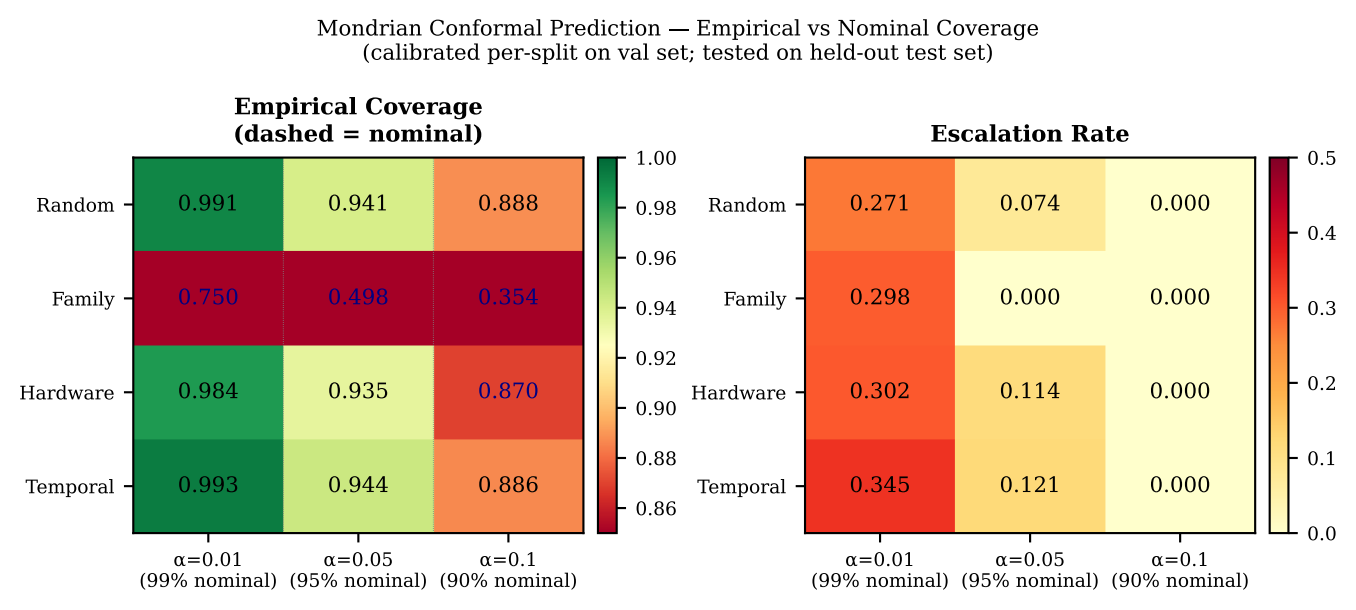
Mondrian Conformal Prediction — Empirical vs Nominal Coverage
(calibrated per-split on val set; tested on held-out test set)
Empirical Coverage
(dashed = nominal)
Random
Family
Hardware
Temporal
0.991
0.941
0.888
0.750
0.498
0.354
0.984
0.935
0.870
0.993
0.944
0.886
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)
Escalation Rate
Random
Family
Hardware
Temporal
0.271
0.074
0.000
0.298
0.000
0.000
0.302
0.114
0.000
0.345
0.121
0.000
0.5
0.4
0.3
0.2
0.1
0.0
α=0.01
(99% nominal)
α=0.05
(95% nominal)
α=0.1
(90% nominal)